

Jorge Novoa Contreras

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Research Interests

Nonlinear Analysis, differential equations, periodic boundary value problems, topological methods, variational methods, monotone operators.

Education

Universidad de Chile	Santiago, Chile
M.Sc. in Applied Mathematics	2025 – 2026
Final grade: 7.0/7.0	
Thesis: <i>Periodic Solutions for Nonlinear Differential Equations in Banach Spaces with Generalized Variable-Exponent Operators</i>	
Advisor: Prof. Raúl Manásevich	
Mathematical Civil Engineering	2020 – 2025
Final grade: 7.0/7.0	
B.Sc. in Engineering Sciences, major in Mathematics	2020 – 2024

Publications

1. Raúl Manásevich and Jorge Novoa, *Global continuum of solutions for systems of ODEs with periodic boundary conditions and generalized variable-exponent operators*, *Communications on Pure and Applied Analysis* **30** (2026), 151–166.

Research Experience

Universidad de Chile	Santiago, Chile
Research Intern	Dec. 2024 – Feb. 2025
• Studied nonlinear differential equations, with emphasis on the existence of periodic solutions for systems involving ϕ-Laplacian-type and variable-exponent operators, under the supervision of Prof. Raúl Manásevich.	

Teaching Experience

Universidad de Chile	Santiago, Chile
Assistant Instructor	2022 – 2026
• Conducted problem-solving sessions and tutoring for: – <i>Multivariable Calculus.</i> – <i>Ordinary Differential Equations.</i> – <i>Laboratory: Enhancing Your Mathematical Skills.</i> – <i>Nonlinear Analysis.</i>	
Teaching Assistant	2022 – 2025
• Graded quizzes, tests, and examinations for: – <i>Multivariable Calculus.</i> – <i>Differential and Integral Calculus.</i>	

- *Complex Variables and Special Functions.*
- *Linear Algebra.*
- *Advanced Calculus and Applications.*

Additional Experience

Center for Mathematical Modeling (CMM)

Santiago, Chile

Selection Reviewer, International Mathematical Modeling Challenge (IMMC)

Nov. 2023 – Dec. 2023

- Reviewed and evaluated mathematical modeling submissions as part of the national selection process for the teams representing Chile at the IMMC.

RialStat

Santiago, Chile

Professional Intern

Jan. 2023 – Feb. 2023; Dec. 2023 – Feb. 2024

- Reviewed and validated economic and financial datasets and supported the development of statistical and macroeconomic models for short-term forecasting of key indicators of the Chilean economy.
- Conducted research and applied statistical and machine-learning methods to support investment portfolio construction, with emphasis on asset prediction and selection.

Languages

- **Spanish:** Native speaker.
- **English:** C1 level (IELTS Academic, overall band score: 7.5).